

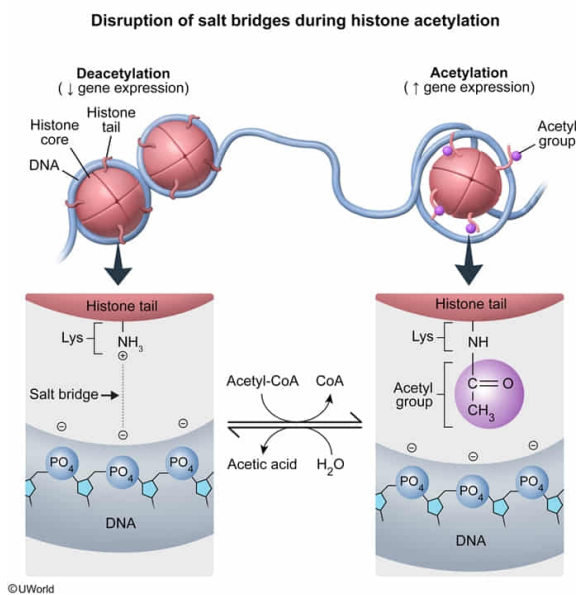
Acetylation of lysine residues in histones increases gene expression because:

- ☐ A. DNA is tightly bound to negatively charged amino acids on histones. [20%]
- ☐ B. the carboxyl oxygen atoms in acetyl groups form hydrogen bonds with nitrogenous bases. [15%]
- ✓ ☒ C. the salt bridges between charged amino acids and phosphate groups are disrupted. [49%]
- ☐ D. lysine residues in histones associate with positively charged phosphate groups in DNA. [14%]

Correct

49% answered correctly

Explanation:



DNA winds tightly around proteins referred to as **histones** to form structural units known as **nucleosomes**. Gene expression depends partially on the **association of histones** with DNA. Genes that are actively transcribed are found on unwound stretches of chromatin called euchromatin, which transcription machinery can easily access. Inactive genes are usually in highly condensed DNA known as heterochromatin, which is much less accessible. Covalent post-translational modifications such as methylation, acetylation, and phosphorylation alter the association of histones with DNA.

DNA is predominantly **negatively charged** due to the **phosphate groups** on the backbone. Histones associate with DNA by forming **salt bridges** between positively charged amino acid residues and negatively charged phosphate groups. These ionic interactions allow histones to bind tightly to DNA and prevent genes from being transcribed. **Acetylation** of histones involves the transfer of acetyl groups from acetyl coenzyme A to positively charged amino groups on lysine or arginine residues. This modification disrupts salt bridges by reducing the positive charge on histones, which allows DNA to unwind and become more accessible to transcription machinery. As a

result, the acetylation of histones causes nucleosomes to relax and **increases gene expression**.

(Choice A) Negatively charged amino acids (**aspartate and glutamate**) would be *repelled* by the negatively charged phosphate groups on DNA.

(Choice B) The carboxyl oxygen atoms in the acetyl groups form **hydrogen bonds** with water rather than the nitrogenous bases on DNA. Acetylation neutralizes the positive charge on histones and decreases their interaction with DNA.

(Choice D) **Lysine** residues are positively charged, and associate with *negatively* charged phosphate groups in DNA. If phosphate groups were positively charged, they would repel lysine.

Educational objective:

Histones associate with DNA by forming salt bridges between their positively charged residues and negatively charged phosphate groups on DNA. Acetylation of histones involves the transfer of an acetyl group to positively charged amino groups on lysine or arginine residues, increasing gene expression by disrupting salt bridges between histones and DNA.

Time spent:0 sec

Subject:
Biochemistry

QID:401351

System:
Carbohydrates, Nucleotides, and Lipids